

Pawara Induwara Ihalavithana

Full-Stack & AI/ML Engineer

Final-year Computer Science undergraduate

CONTACT

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PORTFOLIO
induwara-ihalavithana.vercel.app

GITHUB
github.com/InduwaraIhalavithana

LOCATION
Sri Lanka

TECHNICAL SKILLS

Languages
Python • TypeScript • JavaScript • Java • SQL • Bash

AI / ML
XGBoost • scikit-learn • TensorFlow • PyTorch • pandas • NumPy • Isolation Forest • Autoencoders • Explainable AI

Web & Backend
React • Next.js • Tailwind CSS • FastAPI • SQLAlchemy • Alembic • PostgreSQL • REST APIs • JWT

DevOps & Cloud
Docker • nginx • GitHub Actions CI/CD • Azure VMs • Vercel • Ansible • Linux (Ubuntu Server)

PROFILE

Final-year Computer Science and Technology undergraduate who designs, builds and deploys complete software products for real problems in Sri Lanka. I have shipped three **full-stack platforms** end to end, covering the data model, REST API, React frontend, role-based dashboards, automated tests, CI/CD and cloud deployment, each one usable in English, Sinhala and Tamil. I pair web engineering with **machine learning**, from an explainable XGBoost crop recommender to ongoing research on AI-based counterfeit drug detection.

I became an engineer to help people, so I look for problems that affect daily life and build the software that solves them.

EDUCATION

BSc (Hons) in Computer Science and Technology — Expected 2027

Uva Wellassa University of Sri Lanka · Faculty of Applied Sciences, Badulla

Final year · Graduating 2027

Key coursework: Machine Learning · Parallel & Distributed Computing · Computer Systems Architecture · Databases · Software Engineering

RESEARCH

AI-Enhanced Blockchain Framework for Pharmaceutical Drug Transaction Traceability

Final-year research · Group project · Ongoing · Poster & proposal presented

M.R.T. Gimhan, **P.I. Ihalavithana**, N. Farook · Supervisor: Prof. E.M.U.W.J.B. Ekanayake · AIDICA Lab, Uva Wellassa University

- **Problem.** Sri Lanka's state drug supply chain (NMRA → SPC → MSD → hospitals) depends on paper records that can be forged; in 2023 about 22,500 substandard immunoglobulin vials reached state hospitals.
- **Traceability.** A permissioned blockchain on **Hyperledger Fabric** with **GS1** identifiers gives every pack a tamper-proof record across the whole chain.
- **Detection (my role, AI lead).** I build, tune and validate the **Isolation Forest** and **Autoencoder** models that flag suspicious and counterfeit transactions, with risk scores and real-time alerts stored on-chain.
- **Evaluation.** Models are measured on accuracy, precision, recall and F1; the network on throughput, latency and scalability with **Hyperledger Caliper**.

LABS & OTHER WORK

github.com/InduwaraIhalavithana/Network-Automation-Lab · induwara-ihalavithana.vercel.app

- **Network Automation Lab.** Built two Ubuntu Server VMs in VirtualBox, set a static IP by hand with Netplan, then automated the same change with one **Ansible** playbook over SSH.
- **Portfolio website. Next.js 16**, TypeScript, Tailwind CSS, shadcn/ui and Motion, with dark and light themes and full English, Sinhala and Tamil versions, deployed on Vercel.

MORE SKILLS

Blockchain & Tools

Hyperledger Fabric • Hyperledger Caliper • Git • LaTeX

Quality

pytest • Vitest • Testing Library • ESLint • Ruff • i18n (English / Sinhala / Tamil)

STRENGTHS

- Takes a product from idea to live deployment
- Writes tests and CI from the start
- Designs for local users: trilingual, accessible, mobile-first
- Team research collaboration

EXTRACURRICULAR

Volunteer
IEEE Student Branch, Uva Wellassa University

Volunteer
Microsoft Club, Uva Wellassa University

Member
University Tennis Team, Uva Wellassa University

LANGUAGES

Sinhala
Native

English
Professional working proficiency

INTERESTS

- AI implementation
- AI hardware and emerging chip technologies
- Competitive tennis
- Self-development and motivational speaking

References available upon request.

PROJECTS

SmartAgri: AI-powered agriculture platform

Team of 4 · 2026

Crop advice, farm tracking and a marketplace for Sri Lankan farmers and traders

github.com/Ravindu08/SmartAgri

- Built the **Trader module** and the **Admin dashboard** (user management, reports, CSV import, harvest forecasting) in a three-role system with multi-role accounts.
- Recommends crops with an **XGBoost** model that turns 11 soil, climate and location inputs into explainable results, alongside 7-day weather for all 25 districts.
- Covers the full marketplace order lifecycle: negotiation, confirmation, simulated checkout, delivery and seller ratings.
- Runs on Docker, nginx and HTTPS on an **Azure** VM; GitHub Actions CI/CD publishes images to GitHub Container Registry.

STACK React · FastAPI · PostgreSQL · XGBoost · scikit-learn · Docker · nginx · GitHub Actions · Azure

Sharing Excess: food redistribution platform

Final-year project · 2026

Connects food donors with nearby recipients and NGOs

github.com/Induwaralhalavithana/Sharing-Excess

- Donors post surplus food, people nearby see it first and can request all or part of it; stock is held until the donor responds.
- Designed Donor, Recipient, NGO and Admin roles, with NGO events, ratings, a map view and guided tours.
- Delivers live notifications over **Server-Sent Events**; ships as an installable PWA, responsive down to 320 px.
- Backed by **243 automated tests** (187 pytest, 56 Vitest), including a check that every UI string exists in all three languages.

STACK React 19 · TypeScript · TanStack Query · Zod · Leaflet · FastAPI · SQLAlchemy · PostgreSQL · Docker · GitHub Actions

ConferenceHub: conference management platform

Individual · 2026

Plan, publish and run conferences end to end

github.com/Induwaralhalavithana/Conference-Manager

- Event editor with live preview: cover image, agenda, speakers and an OpenStreetMap venue pin.
- Attendees register without an account, confirm with a 6-digit email code and receive a QR ticket; a waitlist fills freed seats automatically.
- Camera-based check-in desk with a live counter, plus an analytics dashboard with a 14-day trend.
- CI runs Ruff, migration up/down checks on PostgreSQL, pytest, ESLint, TypeScript, Vitest and a production build on every push.

STACK React 18 · TypeScript · Vite · Leaflet · FastAPI · Alembic · PostgreSQL · Gmail SMTP · Docker · GitHub Actions